

Electronics & Communication Systems

Course Introduction



Ing. Pietro Nannipieri – pietro.nannipieri@unipi.it

Prof. Luca Fanucci – luca.fanucci@unipi.it



Pietro Nannipieri

- Digital Systems Architectures, VHDL
- (20 ore)
- Phone: +39 050 2217 660
- Email: pietro.nannipieri@unipi.it

Tutti i team



1012I 22/23 - MICROELECTRONI...

...

Pagina iniziale

Blocco appunti per la classe

Attività

Voti

Reflect

Insights

Canali

Generale

 **Generale**

Post

File

+

+ Nuovo

Carica

Sincronizza

Apri in SharePoint

Documenti > General > **Materiale del corso**

 Nome	Data/ora modifica	Modificato da	+ Aggiungi colonna
 Lecture Slides	Circa un minuto fa	Pietro Nannipieri	
 VHDL & Hands-on Exercises	Circa un minuto fa	Pietro Nannipieri	

Course Material

Slides, VHDL Code from exercise, Lesson recordings

Università di Pisa – ES2025 – Fanucci – Nannipieri

3

Tools

PC-based VHDL simulation tools:

- Aldec (<http://www.aldec.com>): Active HDL
- https://www.aldec.com/en/products/fpga_simulation/active_hdl_student
- **Mentor** (<http://www.mentor.com>): ModelSim
- <http://ghdl.free.fr/> (+ <http://gtkwave.sourceforge.net/>)

Xilinx Development tool:

- <http://www.xilinx.com/products/design-tools/vivado/vivado-webpack.html>

Mentor[®]

A Siemens Business

AMD 

XILINX

OVERVIEW

ModelSim

ModelSim simulates behavioral, RTL, and gate-level code - delivering increased design quality and debug productivity with platform-independent compile. Single Kernel Simulator technology enables transparent mixing of VHDL and Verilog in one design.

Watch webinar

View fact sheet

Get in touch with our sales team 1-800-547-3000



Tools – VHDL Simulation

<https://eda.sw.siemens.com/en-US/ic/modelsim/>

Student edition Currently not available

Questa*-Intel® FPGAs Pro Edition Software Version 24.2

ID	Date	Software Type	Software Package	Version	Operating Systems
826872	7/8/2024	Simulation Tools	Questa*-Intel® FPGAs Pro	24.2 (Latest)	Windows, Linux

The Questa*-Intel® FPGA Pro Edition, Version 24.2 includes functional and security updates. Users should keep their software up-to-date and follow the [technical recommendations](#) to help improve security. Additional security updates are planned and will be provided as they become available. Users should promptly install the latest version upon release.

The Questa*-Intel® FPGA Pro Edition, Version 24.2 is subject to removal from the web when support for all devices in this release are available in a newer version, or all devices supported by this version are obsolete. If you would like to receive customer notifications by e-mail, please subscribe to our [subscribe to our customer notification mailing list](#).

Downloads

[Linux Software](#)

[Windows Software](#)

Questa Software

Questa*-Intel® FPGA and Starter Editions

Download
QuestaSetup-24.2.0.40-windows.exe

Size: 2.2 GB

SHA1: 9d67fbaee4f76d81d550684993a16ea816858511

** Installation size: 9.83 GB

Tools – VHDL Simulation

[LINK](#) 2024 Windows

[LINK](#) 2024 Unix

[LINK](#) 2020 Windows
(smaller installation size)

[LINK](#) 2020 Unix
(smaller installation size)

Tool We will use in class

ModelSim-Intel® FPGA
Edition (includes Starter
Edition)

2020 suggested edition
(plug-in license)

Modelsim

The image shows the Intel FPGA Software Download Center interface. The search bar at the top left contains the text "quartus lite 20" and is highlighted with a red box. Below the search bar, the left sidebar shows filter options for "Intel® Quartus Software" and "Additional Software". The main content area displays "4 Results" and a list of software packages. A red arrow points from the search results to a detailed "Downloads" page on the right.

FPGA Software Download Center

Search: quartus lite 20 (4 Results)

Filter by

- Intel® Quartus Software
- Intel® Quartus® Prime Design Software (4)
- Additional Software
- Intel® FPGA Programming Software (4)
- Intel® FPGA Simulation Tools (4)

Downloads

Multiple Download | Individual Files | Additional Software | Copyleft Licenses

Intel® Quartus® Software

Software Package	Download Link	Size	SHA1
ModelSim-Intel® FPGA Edition (includes Starter Edition)	Download ModelSimSetup-20.1.1.720-windows.exe	1.2 GB	d484e4c7
Intel® Quartus® Prime (includes Nios® II EDS)	Download QuartusLiteSetup-20.1.1.720-windows.exe	1.6 GB	5edd76fc

** Nios® II EDS on Windows requires Ubuntu 18.04 LTS on Windows Subsystem for Linux
** Nios® II EDS requires you to install an Eclipse IDE manually.

<https://fpgasoftware.intel.com/>

PRODUCTS

FPGA Simulation

Active-HDL

Functional Verification

Riviera-PRO

ALINT-PRO

Emulation & Prototyping

HES-DVM

HES Proto-AXI

HES Boards

TySOM Boards

Daughter Cards

RTAX/RTSX Adaptor Boards

RTAX/RTSX Netlist Converter

Cloud

HES-DVM Proto Cloud Edition

Embedded

TySOM™ EDK

Requirements Management

Spec-TRACER

Mil/Aero Verification

DO-254/CTS

University Programs

VIP/IP Products

Active-HDL™ | Student Edition

Free Download

Free Active-HDL Student Edition

Active-HDL Student Edition is a mixed language design entry and simulation tool offered at no cost by Aldec for students to use during their course work.

Licensing

Active-HDL Student Edition includes a "load and go" license. This means students can begin using it immediately after installing.

Key Features of Active-HDL Student Edition

- Mixed language simulator
- Multi-FPGA & EDA Tool Design Flow Manager
- Graphical Design entry & editing
- Code2Graphics and Graphics2Code
- Pre-compiled FPGA vendor libraries
- IEEE Language Support: VHDL, Verilog, SystemVerilog(Design), SystemC
- Waveform Viewer and List Viewer
- Interface with MATLAB®/Simulink®
- HTML and PDF Design Documentation

Tools – VHDL Simulation

- [LINK](#)
- Low Cost Alternative Solution

Vivado Design Suite Evaluation and WebPACK

Evaluate the Vivado Design Suite HLx Edition - Free for 30 days!

If you're looking at Xilinx for the first time or considering Vivado® Design Suite for your design environment, a free 30-day evaluation license gets you started quickly!

- [Download](#) the Vivado Design Suite HLx Edition and start your evaluation today.
- [Get a free 30-day](#) Vivado Design Suite HL System Edition Evaluation License.

Download Vivado HL WebPACK Edition

The Vivado Design Suite HL WebPACK™ Edition is the FREE version of the revolutionary design suite. Vivado HL WebPACK delivers

Vivado Design Suite Evaluation and WebPACK

- [Downloads](#)
- [Vivado Video Tutorials](#)
- [Intellectual Property](#)
- [Support and Documentation](#)
- [EDA Solutions Partners](#)

[+ More](#)

Tools

VHDL Logic synthesis and FPGA Programming ([LINK](#))

Vivado

The image shows the Xilinx Vivado Downloads page. The main page has a 'Downloads' header and a navigation bar with links like 'Log4j Vulnerability Patch', 'Licensing Help', and 'Alveo'. Below this, there are tabs for 'Vivado (HW Developer)', 'Vitis (SW Developer)', 'Vitis Embedded Platforms', 'Alveo Packages', and 'PetaLinux'. The 'Vivado (HW Developer)' tab is selected. Under this tab, there is a 'Version' section with a table showing releases from 2021.2 down to 2020.2. The 'Vivado Archive' link is highlighted with a red box. Below the version table, there is a section for 'Vivado Design Suite - HLx Editions - 2020.2 Full Product Installation' with an 'Important Information' link and a 'Download' button. A modal window is open on the right, showing details for the 'Xilinx Unified Installer 2020.2: Windows Self Extracting Web Installer (EXE - 248.44 MB)'. A red arrow points to the download link in the modal. The modal also displays the MD5 SUM Value and a 'Download Verification' section with buttons for 'Digests', 'Signature', and 'Public Key'. Below this, there is a section for the 'Linux Self Extracting Web Installer (BIN - 354.08 MB)' with similar verification options.

Downloads

Log4j Vulnerability Patch Licensing Help Alveo

Vivado (HW Developer) Vitis (SW Developer) Vitis Embedded Platforms Alveo Packages PetaLinux

Version We strongly recommend using the latest releases available.

Version	Release
2021.2	2020
2021.1	2020.2
2020.2	2020.2

Vivado Archive

CAE Vendor Libraries Archive

Vivado Design Suite - HLx Editions - 2020.2 Full Product Installation

Important Information Download

Xilinx Unified Installer 2020.2: Windows Self Extracting Web Installer (EXE - 248.44 MB)

MD5 SUM Value : 102bb67c6806a6667dc7176be7997475

Download Verification ⓘ

Digests Signature Public Key

Xilinx Unified Installer 2020.2: Linux Self Extracting Web Installer (BIN - 354.08 MB)

MD5 SUM Value : 0c74a74cbef649dceea34774c5bca490

Download Verification ⓘ

Digests Signature Public Key

<https://www.xilinx.com/support/download.html>

Vivado



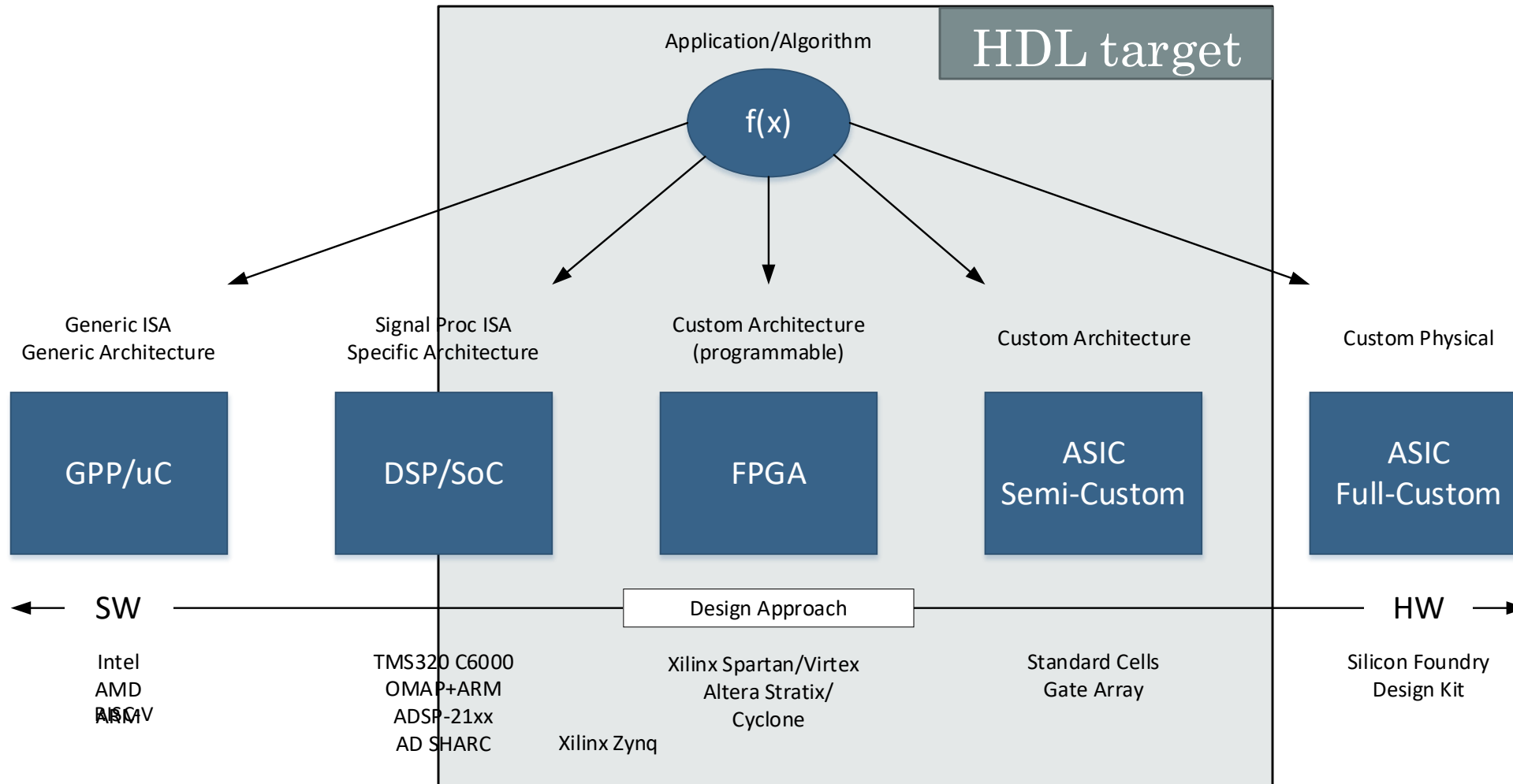
Free Version

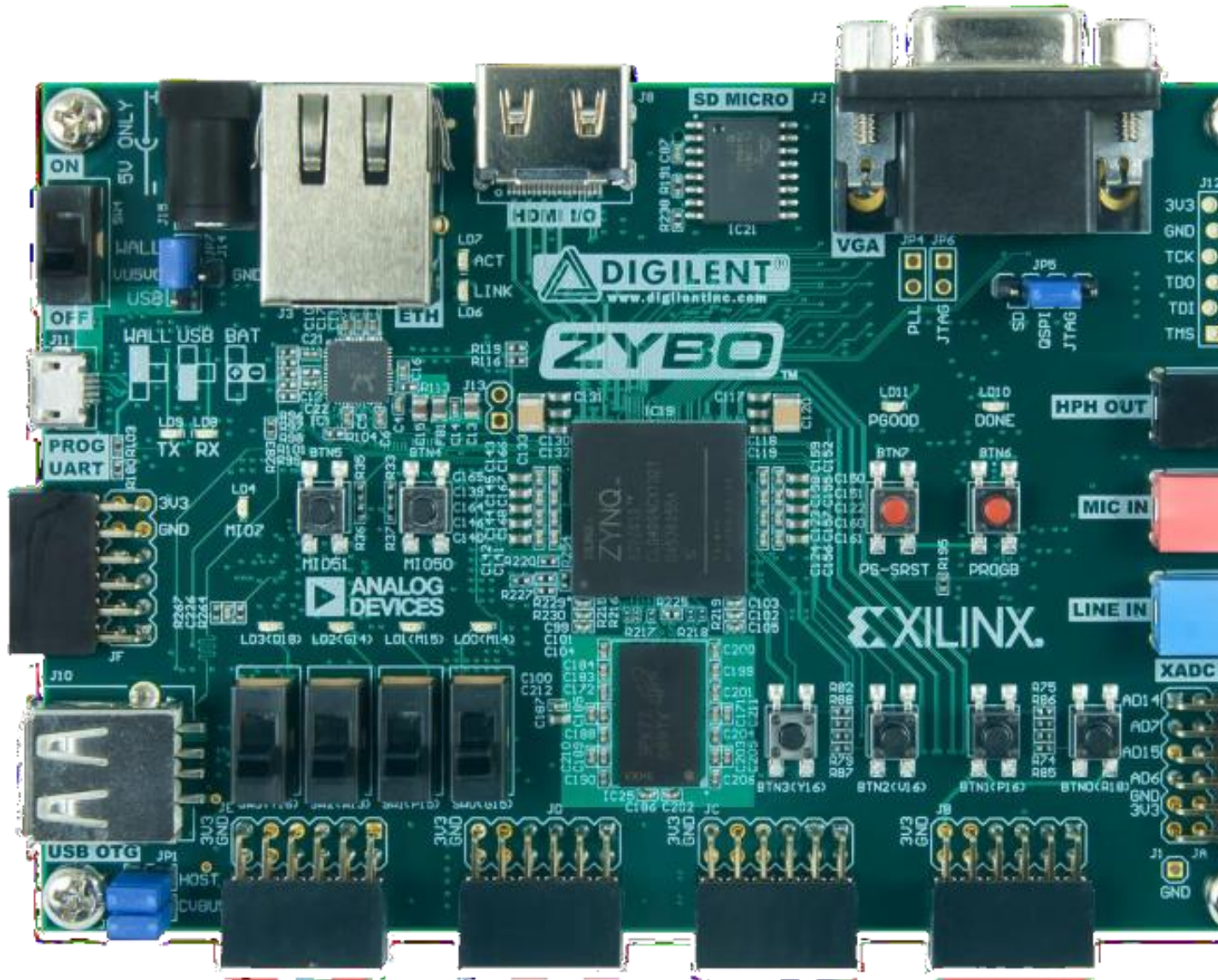


To save some space

Digital Design Technology Map:

From a $f(x)$ to *Hardware Acceleration*





Hands-on:

VHDL modelling, logic
synthesis and FPGA
Programming

During exercitation we will
design from scratch a
complete digital system, from
requirements to the
implementation on a FPGA
Board and its test.

Hands-on: DDFS System Description

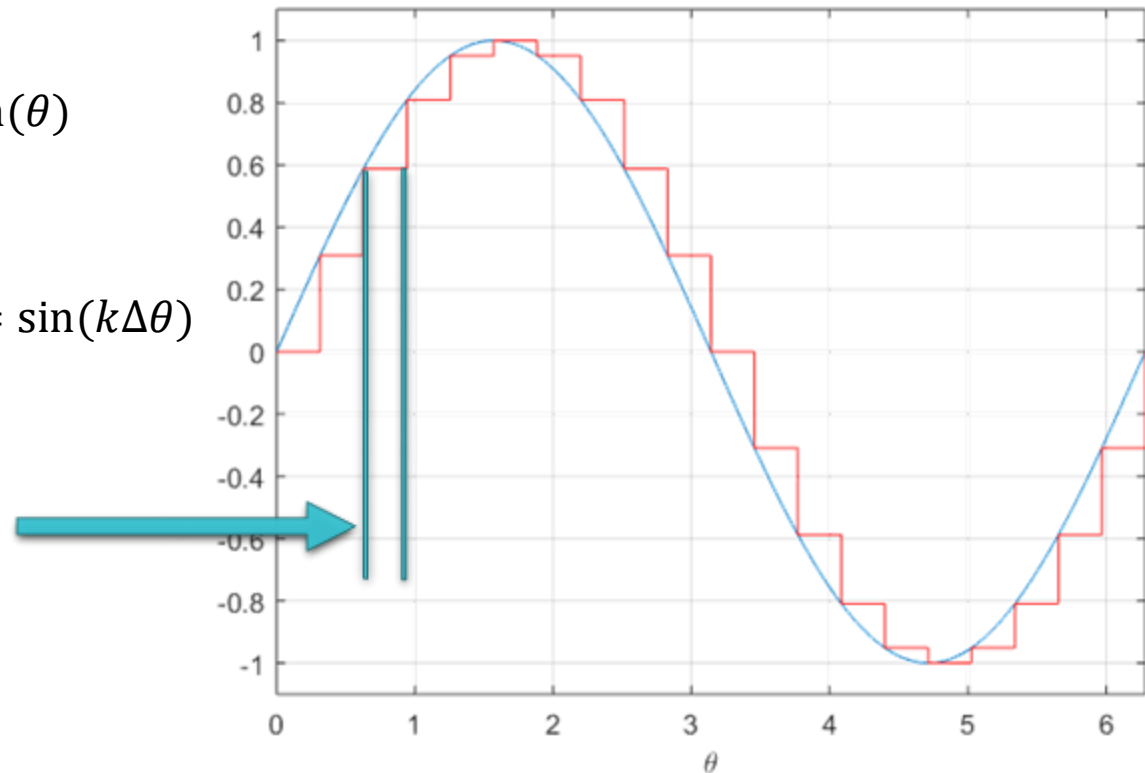
The DDFS (Direct Digital Frequency Synthesizer) is a digital system able to generate a quantized *sin* waveform, with frequency control.

Building block of many Telecommunication systems

Suppose we want to generate a single tone at frequency f :

- Ideal output: $\sin(2\pi f t) = \sin(\theta)$
- Discrete time: $\sin(2\pi f k T_{ck}) = \sin(k\Delta\theta)$
- Phase resolution:
 - Depends on f
 - Depends on f_{ck}

$$\Delta\theta = 2\pi \frac{f}{f_{ck}}$$



Exam

- Oral test on topics covered in the course
- Digital Design Project (Optional BUT Recommended)
 - (VHDL and automatic logic synthesis for FPGA technology)
 - Realization of digital electronic system of similar complexity as seen during class hands-on exercises
 - 1 - 2 people
 - Request for project assignment via email
 - Email delivery of source code and testbench together with the technical report
 - Design Review/Discussion before the oral exam

Questions?

